

# Newman-Penrose Formalism Calculations

Stephani et al., Equation (15.87)

## Metric

$$ds^2 = -dt \otimes dt + e^{(-bt^2 + bz^2 + c_2 t + c_1 z)} dx \otimes dx + e^{(-bt^2 + bz^2 + c_2 t + c_1 z)} dy \otimes dy + dz \otimes dz$$

## Metric Functions

$$P = e^{(bz^2 + c_1 z)}$$

$$Q = e^{(-bt^2 + c_2 t)}$$

## Coordinates

$$(x^\mu) = (t, x, y, z)$$

## Null Tetrad

*Covariant components*

$$l = \left(\frac{1}{2} \sqrt{2}\right) dt + \left(\frac{1}{2} \sqrt{2}\right) dz$$

$$n = \left(\frac{1}{2} \sqrt{2}\right) dt + \left(-\frac{1}{2} \sqrt{2}\right) dz$$

$$m = \left(-\frac{1}{2} \sqrt{2} e^{(-\frac{1}{2} bt^2 + \frac{1}{2} bz^2 + \frac{1}{2} c_2 t + \frac{1}{2} c_1 z)}\right) dx + \left(\frac{1}{2} i \sqrt{2} e^{(-\frac{1}{2} bt^2 + \frac{1}{2} bz^2 + \frac{1}{2} c_2 t + \frac{1}{2} c_1 z)}\right) dy$$

$$\bar{m} = \left(-\frac{1}{2} \sqrt{2} e^{(-\frac{1}{2} bt^2 + \frac{1}{2} bz^2 + \frac{1}{2} c_2 t + \frac{1}{2} c_1 z)}\right) dx + \left(-\frac{1}{2} i \sqrt{2} e^{(-\frac{1}{2} bt^2 + \frac{1}{2} bz^2 + \frac{1}{2} c_2 t + \frac{1}{2} c_1 z)}\right) dy$$

*Contravariant components*

$$l = \left(-\frac{1}{2}\sqrt{2}\right) dt + \left(\frac{1}{2}\sqrt{2}\right) dz$$

$$n = \left(-\frac{1}{2}\sqrt{2}\right) dt + \left(-\frac{1}{2}\sqrt{2}\right) dz$$

$$m = \left(-\frac{1}{2}\sqrt{2}e^{\left(\frac{1}{2}bt^2 - \frac{1}{2}bz^2 - \frac{1}{2}c_2t - \frac{1}{2}c_1z\right)}\right) dx + \left(\frac{1}{2}i\sqrt{2}e^{\left(\frac{1}{2}bt^2 - \frac{1}{2}bz^2 - \frac{1}{2}c_2t - \frac{1}{2}c_1z\right)}\right) dy$$

$$\overline{m} = \left(-\frac{1}{2}\sqrt{2}e^{\left(\frac{1}{2}bt^2 - \frac{1}{2}bz^2 - \frac{1}{2}c_2t - \frac{1}{2}c_1z\right)}\right) dx + \left(-\frac{1}{2}i\sqrt{2}e^{\left(\frac{1}{2}bt^2 - \frac{1}{2}bz^2 - \frac{1}{2}c_2t - \frac{1}{2}c_1z\right)}\right) dy$$

## Directional Derivatives

$$D = -\frac{1}{2}\sqrt{2}\partial_t + \frac{1}{2}\sqrt{2}\partial_z$$

$$\Delta = -\frac{1}{2}\sqrt{2}\partial_t - \frac{1}{2}\sqrt{2}\partial_z$$

$$\delta = -\frac{1}{2}\sqrt{2}e^{\left(\frac{1}{2}bt^2 - \frac{1}{2}bz^2 - \frac{1}{2}c_2t - \frac{1}{2}c_1z\right)}\partial_x + \frac{1}{2}i\sqrt{2}e^{\left(\frac{1}{2}bt^2 - \frac{1}{2}bz^2 - \frac{1}{2}c_2t - \frac{1}{2}c_1z\right)}\partial_y$$

$$\overline{\delta} = -\frac{1}{2}\sqrt{2}e^{\left(\frac{1}{2}bt^2 - \frac{1}{2}bz^2 - \frac{1}{2}c_2t - \frac{1}{2}c_1z\right)}\partial_x - \frac{1}{2}i\sqrt{2}e^{\left(\frac{1}{2}bt^2 - \frac{1}{2}bz^2 - \frac{1}{2}c_2t - \frac{1}{2}c_1z\right)}\partial_y$$

## Spin Coefficients

$$\rho = -\frac{1}{4}\sqrt{2}(2bt + 2bz + c_1 - c_2)$$

$$\sigma = 0$$

$$\kappa = 0$$

$$\mu = \frac{1}{4} \sqrt{2} (2bt - 2bz - c_1 - c_2)$$

$$\lambda = 0$$

$$\nu = 0$$

$$\alpha = 0$$

$$\beta = 0$$

$$\tau = 0$$

$$\pi = 0$$

$$\epsilon = 0$$

$$\gamma = 0$$

## Ricci Tensor Components

$$\Phi_{00} = -\frac{1}{8} (2bt + 2bz + c_1 - c_2)^2$$

$$\Phi_{01} = 0$$

$$\Phi_{02} = 0$$

$$\Phi_{10} = 0$$

$$\Phi_{11} = \frac{1}{16} (2bt + 2bz + c_1 - c_2)(2bt - 2bz - c_1 - c_2)$$

$$\Phi_{12} = 0$$

$$\Phi_{20} = 0$$

$$\Phi_{21} = 0$$

$$\Phi_{22} = -\frac{1}{8} (2bt - 2bz - c_1 - c_2)^2$$

$$\Lambda = \frac{1}{4} b^2 t^2 - \frac{1}{4} b^2 z^2 - \frac{1}{4} b c_2 t - \frac{1}{4} b c_1 z - \frac{1}{16} c_1^2 + \frac{1}{16} c_2^2 - \frac{1}{3} b$$

## Weyl Tensor Components

$$\Psi_0 = 0$$

$$\Psi_1 = 0$$

$$\Psi_2 = -\frac{1}{3} b$$

$$\Psi_3 = 0$$

$$\Psi_4 = 0$$

## Newman-Penrose Equations

*Equations are vanished.*

## Bianchi Identities

*Equations are vanished.*

## Petrov Type

Type "*D*"